BE-104 · BTech First Year · UIT, Barkatullah University, Bhopal

Basic Mechanical Engineering — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

UNIT I Boilers & Thermodynamics of Steam

1. Boiler – Definition & Principal Parts

DEFINITION **Boiler (steam generator)** ek closed vessel hai jisme fuel ke combustion se mili heat se water ko **steam** mein convert kiya jata hai, specified pressure aur temperature par (IBR Act ke according capacity > 22.75 litre).

Shell: cylindrical steel vessel, water + steam rakhta hai.

Furnace/Firebox & Grate: fuel jalane ki jagah; grate par solid fuel.

Fire tubes / Water tubes / Flue: hot gases aur water ke beech heat transfer path.

Water space & Steam space: neeche water, upar steam collect.

Chimney: draught paida karna, flue gases bahar. **Ash pit:** ash collect.

Brick/Refractory setting: gas passage banana, heat loss kam karna.

Classification: Fire-tube (gases tubes ke andar, water bahar) vs Water-tube (water tubes ke andar); internally vs externally fired; horizontal / vertical / inclined; stationary vs mobile; natural vs forced circulation; low / high pressure.

2. Important Boilers

Feature	Cochran	Locomotive	Lancashire	Babcock & Wilcox
Type	Fire tube, vertical, multi-tubular	Fire tube, horizontal, multi-tubular, mobile	Fire tube, horizontal, stationary, 2 flue tubes	Water tube, horizontal drum, inclined tubes, stationary
Firing	Internally fired	Internally fired	Internally fired	Externally fired
Draught	Natural	Artificial (exhaust steam blast pipe)	Natural	Natural
Pressure (approx)	up to ~6.5 bar	~14 bar	~15–16 bar	up to ~40 bar
Capacity (approx)	~3500 kg/h	~7000–9000 kg/h	~9000 kg/h	~40,000 kg/h
Use	Small industries, compact	Railway engines	Sugar mills, textile, process steam	Power plants

Cochran Boiler

- Vertical cylindrical shell, **hemispherical crown** aur hemispherical firebox (max strength, min radiation loss).
- Gases firebox → **combustion chamber** → horizontal smoke (fire) tubes → smoke box → chimney.
- Kam floor area, portable, kisi bhi fuel ke liye suitable.

Locomotive Boiler

- Firebox → bahut saare fire tubes → smoke box → chimney. Steam dome se steam li jati hai, superheater bhi hota hai.
- Draught: engine ka **exhaust steam blast pipe** se chimney mein chhoda jata hai → forced draught. High steaming rate, compact.

Lancashire Boiler

- Shell dia 2–3 m, length 7–9 m; **do bade flue (fire) tubes** jinke front end par grate.
- Gas path: flue tubes (front→back) → **bottom flue** (back→front) → do **side flues** (front→back) → main flue → chimney. Is three-pass se heat transfer badhta hai.
- Galloway tubes / dampers se draught control. Large water space → load fluctuation easily handle.

Babcock & Wilcox Boiler

- Horizontal steam drum + water tubes ~10–15° incline par, uptake & downtake **headers** se connected.
- Natural circulation: hot tubes mein water halka hokar uptake header se drum mein, thanda water downtake se wapas → continuous circulation. Baffles gases ko zig-zag path dete hain. Superheater tubes drum ke neeche.

3. Boiler Mountings & Accessories

YAAD RAKHO **Mountings** = safety aur control ke liye, boiler par **compulsory**. **Accessories** = efficiency badhane ke liye, **optional**.

Mountings (safety/control):

- **Water level indicator** – water level dikhata hai (2 per boiler).
- **Pressure gauge** – steam pressure (Bourdon tube).

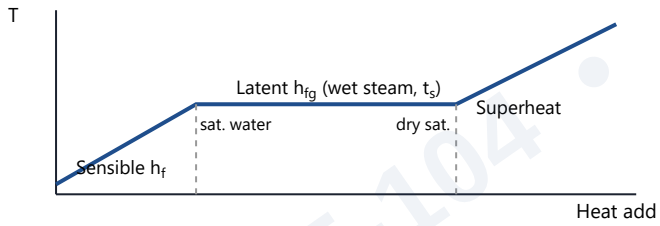
Accessories (efficiency):

- **Economiser** – flue gases se **feed water** preheat.
- **Air preheater** – flue gases se **combustion air** preheat.

- **Safety valve** – limit se zyada pressure par extra steam release (dead weight, lever, spring loaded, high-steam low-water).
- **Fusible plug** – water level girne par low-melting alloy pighalta hai, steam furnace mein jaakar fire bujhati hai.
- **Steam stop valve** – steam flow on/off. **Feed check valve** – feed water andar, wapas nahi (non-return).
- **Blow-off cock** – sediment/mud nikalna. **Manhole** – inspection & cleaning.
- **Superheater** – saturated steam ko same pressure par superheat.
- **Feed pump** – high pressure par water boiler mein bhejna. **Injector** – steam jet se feed water push (small/loco boilers).
- **Steam separator** – steam se water particles alag. **Steam trap** – condensate drain bina steam loss.

4. Thermodynamics of Steam

Water ko constant pressure par heat karo: pehle temp saturation temp t_s tak badhta hai, phir boiling (temp constant), end mein superheating.



Steam formation at constant pressure

Dryness fraction (x): wet steam mein dry steam ka mass fraction. x = 1 dry, x = 0 saturated water.

Sensible heat (h_f): 0 °C water ko t_s tak garam karne ki heat (temp change hota hai).

Latent heat (h_{fg}): t_s par saturated water ko dry saturated steam banane ki heat (temp constant). Pressure badhne par h_{fg} ghatta hai (critical point 221.2 bar par zero).

Wet steam: steam + water droplets. **Dry saturated:** koi droplet nahi, t_s par. **Superheated:** t_s se upar temperature (t_{sup}); **degree of superheat** = $t_{sup} - t_s$.

Internal energy (u): steam mein store energy = enthalpy − external work of evaporation (pv).

FORMULAS (PER KG)

$$x = \frac{m_s}{m_s + m_w} \quad | \quad \text{Wet: } h = h_f + x \cdot h_{fg}, \;\; v \approx x \cdot v_g$$

$$\text{Dry sat.: } h = h_g = h_f + h_{fg} \quad | \quad \text{Superheated: } h = h_g + c_{ps}(t_{sup} - t_s), \;\; c_{ps} \approx 2.1 \text{ kJ/kg·K}$$

$$v_{sup} = v_g \cdot T_{sup}/T_s \quad | \quad \text{Internal energy } u = h - p \cdot v \;\; (p \text{ in kPa, } v \text{ in m}^3\text{/kg} \rightarrow \text{kJ/kg})$$

SOLVED EXAMPLE Steam at **10 bar, x = 0.9**. Steam table: t_s = 179.9 °C, h_f = 763, h_{fg} = 2015, h_g = 2778 kJ/kg, v_g = 0.1944 m³/kg.
h = 763 + 0.9×2015 = **2576.5 kJ/kg**; v = 0.9×0.1944 = 0.175 m³/kg; u = 2576.5 − 1000×0.175 = **2401.5 kJ/kg**.
Agar same pressure par **250 °C superheated**: h = 2778 + 2.1(250 − 179.9) = **2925.2 kJ/kg**.

Determination of Dryness Fraction – Calorimeters

Calorimeter	Principle	Result
Barrel (bucket)	Steam ko known mass ke cold water mein condense karo; heat lost by steam = heat gained by water.	$m_s(h_{f1} + x\,h_{fg} - c_w t_2) = (m_w + m_e)c_w(t_2 - t_1)$. Radiation losses ke karan approximate.
Separating	Mechanical separation: steam ki direction achanak badalne se heavy water droplets alag ho jaate hain.	x = M/(M + m); M = dry steam condensed, m = separated water. Very wet steam ke liye.
Throttling	Steam ko low pressure (≈1 bar) tak throttle karo taki wo superheated ho jaye; $h_1 = h_2$.	$h_{f1} + x\,h_{fg1} = h_{g2} + c_{ps}(t_{sup2} - t_{s2})$. Sirf thoda wet steam (x > ~0.93).
Separating + Throttling	Pehle separator, phir throttle.	**$x = x_1 \times x_2$** — most accurate, kisi bhi wetness ke liye.

NUMERICAL 10 bar steam throttle hokar 1 bar, 115 °C par aati hai. 1 bar: t_s = 99.6 °C, h_g = 2675 kJ/kg. h_2 = 2675 + 2.1(115 − 99.6) = 2707.3. ⇒ x = (2707.3 − 763)/2015 = **0.965**.

Steam Processes

Process	Condition	Work W (kJ/kg)	Heat Q (kJ/kg)
Constant pressure	p = C (boiler, superheater)	$p(v_2 - v_1)$	$h_2 - h_1$
Constant volume	v = C (closed rigid vessel); $x_2 v_{g2} = x_1 v_{g1}$	0	$u_2 - u_1$

Process	Condition	Work W (kJ/kg)	Heat Q (kJ/kg)
Hyperbolic	pv = C (wet steam expansion)	$p_1v_1 \ln(v_2/v_1)$	$(u_2 - u_1) + W$
Throttling	Valve/narrow passage se expansion, no work, no heat	0	0; $\mathbf{h_1 = h_2}$ (isenthalpic), pressure drop, dryness badhti hai

Note: wet steam ke liye isothermal = constant pressure process hota hai; hyperbolic (pv = C) steam mein isothermal nahi hai.

UNIT II I.C. Engines & Laws of Thermodynamics

1. Internal Combustion Engine – Classification

IC engine mein fuel cylinder ke **andar** jalta hai (petrol, diesel engine); **EC engine** mein bahar (steam engine/turbine).

Fuel: petrol, diesel, gas (CNG/LPG), bi-fuel.

Ignition: Spark Ignition (SI) – spark plug; Compression Ignition (CI) – hot compressed air.

Strokes: four-stroke (2 rev/cycle), two-stroke (1 rev/cycle).

Cycle: Otto (const. volume), Diesel (const. pressure), Dual.

Cooling: air / water cooled. **Cylinders:** single/multi – inline, V, radial, opposed.

Application/speed: automobile, marine, stationary, aircraft; low/medium/high speed.

Terms: Bore (D), Stroke (L) = distance TDC→BDC, Swept volume $V_s = (\pi/4)D^2L$, Clearance volume V_c, **Compression ratio $r = (V_s+V_c)/V_c$**.

2. Four-Stroke Petrol (SI) Engine

1. **Suction:** piston TDC→BDC, inlet valve open; **air–petrol mixture** (carburettor/injector se) andar.
2. **Compression:** dono valves band, piston BDC→TDC; mixture compress (r = 6–10).
3. **Power (expansion):** TDC ke paas **spark plug** spark deta hai, mixture jalta hai, pressure badhta hai, piston neeche dhakka khata hai → work milta hai.
4. **Exhaust:** exhaust valve open, piston upar aakar burnt gases bahar nikalta hai.

Ek cycle = 4 strokes = crankshaft ke **2 revolutions**; sirf ek power stroke, isliye heavy **flywheel** chahiye.

3. Four-Stroke Diesel (CI) Engine

Same 4 strokes, par: suction mein **sirf air**; compression ratio high **(14–22)** se air temp ~600–700 °C ho jata hai; compression ke end par **fuel injector** diesel spray karta hai jo **self-ignite** hota hai (koi spark plug nahi). Phir power aur exhaust stroke.

4. Two-Stroke Engines

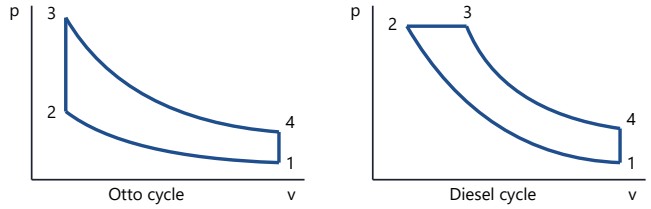
Valves ki jagah cylinder wall mein **ports** (inlet, transfer, exhaust) hote hain jinhe piston khud kholta/band karta hai. Crankcase sealed hota hai aur charge ka pre-compression karta hai. **Har revolution mein ek power stroke.**

- **Upward stroke (BDC→TDC):** piston ke upar charge compress; piston ke neeche crankcase mein partial vacuum → inlet port khulta hai, fresh charge crankcase mein.
- **Downward stroke (TDC→BDC):** ignition → expansion (power); end ke paas pehle **exhaust port** khulta hai (blow-down), phir **transfer port** se crankcase ka compressed charge cylinder mein aata hai aur burnt gases ko bahar dhakelta hai (**scavenging**, deflector-type piston crown).
- **2-stroke petrol:** crankcase mein air-fuel-lubricating oil mixture, spark plug. **2-stroke diesel:** sirf air scavenge hoti hai (blower se), TDC par injector se fuel; badi marine engines.

Point	4-stroke	2-stroke
Power stroke	1 per 2 rev	1 per rev
Valves	Poppet valves + cam	Ports (simple, light)
Flywheel	Heavy	Lighter
η / fuel	Higher η, less fuel	Lower η (charge loss in scavenging)
Power/weight	Less	More (theoretically 2×)
Use	Cars, trucks, buses	Mopeds, small pumps, marine diesel

Point	Petrol (SI)	Diesel (CI)
Cycle	Otto (V = C heat add.)	Diesel (p = C heat add.)
Intake	Air + fuel mixture	Only air
Ignition	Spark plug	Injector, self-ignition
Comp. ratio	6–10	14–22
Weight / cost	Light, cheap	Heavy, costly
Speed / η	High speed, less η	Lower speed, more η

5. Working Cycles – Otto & Diesel



1–2 adiabatic compression, 2–3 heat addition (Otto: V = C; Diesel: p = C), 3–4 adiabatic expansion, 4–1 heat rejection at V = C

AIR-STANDARD EFFICIENCY

$$\eta_{Otto} = 1 - \frac{1}{r^{\gamma-1}}$$

$$\eta_{Diesel} = 1 - \frac{1}{r^{\gamma-1}} \cdot \frac{\rho^{\gamma} - 1}{\gamma(\rho - 1)}$$

r = compression ratio, cut-off ratio $\rho = v_3/v_2$

NUMERICALS (γ = 1.4) Otto, r = 8: $8^{0.4}$ = 2.297 ⇒ η = 1 − 0.435 = **56.5%**. Diesel, r = 16, ρ = 2: $16^{0.4}$ = 3.031; ($2^{1.4}$ − 1)/(1.4×1) = 1.171 ⇒ η = 1 − 1.171/3.031 = **61.4%**. Same r par $\eta_{Otto} > \eta_{Diesel}$, lekin diesel high r par chalta hai isliye practically zyada efficient.

6. Indicated Power, Brake Power & Efficiencies

FORMULAS

$$IP = \frac{p_m \cdot L \cdot A \cdot n \cdot k}{60} \text{ watt}$$

(p_m in N/m², L in m, A in m²; n = N/2 for 4-stroke, n = N for 2-stroke; k = no. of cylinders)

$$BP = \frac{2\pi NT}{60} \text{ watt} \quad | \quad FP = IP - BP \quad | \quad \eta_{mech} = BP/IP$$

$$\eta_{bth} = \frac{BP}{m_f \times CV} \quad | \quad \eta_{ith} = \frac{IP}{m_f \times CV} \quad | \quad \eta_{rel} = \eta_{ith}/\eta_{air\text{-}std} \quad (m_f \text{ in kg/s})$$

p_m = (indicator diagram area × spring constant)/diagram length; BSFC = m_f(kg/h)/BP(kW)

IP = cylinder ke andar gas dwara piston par develop power (indicator diagram se). **BP** = crankshaft par milne wali useful power (brake/dynamometer se). **FP** = friction mein loss.

SOLVED EXAMPLE Single-cylinder 4-stroke diesel: D = 100 mm, L = 150 mm, N = 1500 rpm, p_m = 6 bar, torque T = 45 N·m, fuel 2.2 kg/h, CV = 44,000 kJ/kg.

A = (π/4)(0.1)² = 7.854×10^{-3} m², n = 750. IP = 6×10^5×0.15×7.854×10^{-3}×750/60 = **8.84 kW**.

BP = 2π×1500×45/60 = **7.07 kW**; FP = 1.77 kW; η_{mech} = 7.07/8.84 = **80%**.

Heat supplied = (2.2/3600)×44000 = 26.89 kW ⇒ η_{bth} = 7.07/26.89 = **26.3%**, η_{ith} = 8.84/26.89 = **32.9%**.

7. Thermodynamics – Basic Concepts

System: space/quantity of matter jis par study focus hai; baaki **surroundings**; beech mein **boundary** (fixed ya movable).

- **Closed system:** mass transfer nahi, energy (heat/work) transfer ho sakta hai – piston-cylinder gas.
- **Open system (control volume):** mass + energy dono cross karte hain – turbine, compressor, nozzle.
- **Isolated system:** na mass na energy – universe, ideal thermos flask.

Property: state ki measurable characteristic (p, V, T, U, H, S). **Intensive** – mass par depend nahi (p, T, density); **Extensive** – mass par depend (V, U, H). Property ek **point function** hai; heat aur work **path functions** hain.

State = properties ka set; **Process** = state change; **Cycle** = initial state par wapas aana; **Quasi-static process** = bahut dheere, har point equilibrium par. **Thermodynamic equilibrium** = thermal + mechanical + chemical equilibrium.

Process (ideal gas)	Law	Work done $W_{1\text{-}2}$
Isochoric	V = C, p/T = C	0
Isobaric	p = C, V/T = C	$p(V_2 - V_1)$
Isothermal	T = C, pV = C	$p_1V_1 \ln(V_2/V_1)$
Adiabatic (reversible)	pV^{γ} = C, Q = 0	$(p_1V_1 - p_2V_2)/(\gamma - 1)$
Polytropic	pV^{n} = C	$(p_1V_1 - p_2V_2)/(n - 1)$

8. Laws of Thermodynamics

Law	Statement / Meaning
Zeroth law	Agar body A aur B dono body C ke saath thermal equilibrium mein hain, to A aur B bhi aapas mein thermal equilibrium mein honge. → **Temperature** concept aur **thermometer** ka basis.
First law	Energy na create hoti hai na destroy, sirf form badalti hai. Cycle ke liye $\oint \delta Q = \oint \delta W$. Process ke liye **Q − W = ΔU** (Q added +ve, W by system +ve). Open steady flow: q = Δh + ΔKE + ΔPE + w (SFEE). **PMM-1** (bina energy ke work dene wali machine) impossible.
Second law	**Kelvin–Planck:** koi bhi cyclic heat engine ek hi reservoir se heat lekar poori ki poori work mein convert nahi kar sakta (η < 100%). **Clausius:** heat apne aap thandi body se garam body ki taraf nahi ja sakti bina external work ke (refrigerator ko work chahiye). Dono equivalent hain. **PMM-2** impossible. Process ki direction batata hai; entropy concept.
Third law	Perfect crystalline substance ki **entropy absolute zero (0 K) par zero** hoti hai. Isliye 0 K finite steps mein achieve nahi ho sakta.

HEAT ENGINE

$$\eta = W/Q_1 = 1 - Q_2/Q_1 \quad | \quad \eta_{Carnot} = 1 - T_2/T_1 \text{ (T in K)} \quad | \quad COP_{ref} = Q_2/W$$

NUMERICALS (i) Gas ko 100 kJ heat di, wo 40 kJ work karti hai: ΔU = 100 − 40 = **60 kJ**. (ii) Engine between 527 °C & 27 °C: η_{Carnot} = 1 − 300/800 = **62.5%**. Koi inventor isse zyada η claim kare to wo impossible hai.

UNIT III Heat Transfer & Psychrometry

1. Modes of Heat Transfer

Mode	Mechanism	Law / Formula	Applications
Conduction	Solid/stationary fluid mein molecule-to-molecule (lattice vibration, free electrons); medium chahiye, bulk motion nahi.	Fourier: $Q = -kA\frac{dT}{dx}$; k in W/m·K	Boiler/furnace walls, insulation, cooking pan, engine cylinder wall
Convection	Fluid ki bulk motion se; **natural** (density difference) ya **forced** (fan/pump).	Newton's law of cooling: $Q = hA(T_s - T_\infty)$; h in W/m²·K	Radiators, AC/fan cooling, condensers, boiler tubes
Radiation	Electromagnetic waves; **medium ki zarurat nahi** (vacuum mein bhi).	Stefan–Boltzmann: $Q = \varepsilon\sigma A(T_1^4 - T_2^4)$, $\sigma = 5.67\times10^{-8}$ W/m²K⁴	Sun heat, furnace, solar collector, boiler furnace

Applications overall: heat exchangers (radiator, condenser, evaporator), boilers, IC engine cooling, electronics cooling, building insulation, refrigeration & AC.

2. Unidirectional Steady-State Conduction (Plane Wall)

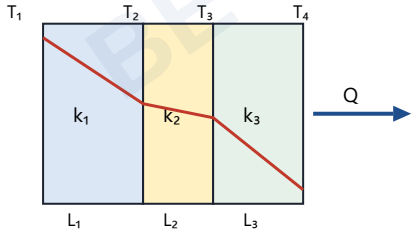
Steady state = temperature time ke saath change nahi hota; **one-dimensional** = temperature sirf ek direction (x) mein vary karta hai. Fourier law ko 0 se L tak integrate karne par (k constant) temperature profile **linear** aata hai:

FORMULA

$$Q = \frac{kA(T_1 - T_2)}{L} = \frac{T_1 - T_2}{R_{th}}, \quad R_{th} = \frac{L}{kA} \text{ (conduction)}, \quad R_{conv} = \frac{1}{hA} \text{ (K/W)}$$

Electrical analogy: Q ↔ current I, ΔT ↔ voltage V, R_{th} ↔ resistance; series walls mein resistances **add** hoti hain.

3. Heat Transfer through Composite Slab



Three layers in series (same Q through each)

COMPOSITE WALL

$$Q = \frac{T_1 - T_4}{\frac{L_1}{k_1A} + \frac{L_2}{k_2A} + \frac{L_3}{k_3A}}$$

$$\text{With fluids both sides: } Q = \frac{T_i - T_o}{1/h_iA + \Sigma L/kA + 1/h_oA}$$

$$\text{Overall: } Q = UA\Delta T, \quad 1/U = 1/h_i + \Sigma L/k + 1/h_o$$

SOLVED EXAMPLE Furnace wall: firebrick L_1 = 0.2 m, k_1 = 0.7 W/mK; insulation L_2 = 0.05 m, k_2 = 0.05 W/mK. Inner face 400 °C, outer 40 °C. Per m^2:
R_1 = 0.2/0.7 = 0.286, R_2 = 0.05/0.05 = 1.0 ⇒ ΣR = 1.286 K/W. Q = (400 − 40)/1.286 = **280 W/m²**.
Interface temp: T_2 = 400 − 280×0.286 = **320 °C**. (Note: thin insulation ka resistance brick se zyada — low k ka kamaal.)

4. Psychrometry – Properties of Moist Air

Psychrometry = atmospheric air (dry air + water vapour) ki properties ka study; AC, cooling towers, drying mein use. Dalton's law: $p = p_a + p_v$ (p_v = vapour partial pressure).

Property	Meaning
Dry Bulb Temp (DBT)	Ordinary thermometer (dry bulb) se measured air temperature.
Wet Bulb Temp (WBT)	Thermometer jiska bulb wet wick se dhaka ho; evaporation cooling ke karan WBT ≤ DBT. **Wet bulb depression** = DBT − WBT.
Dew Point Temp (DPT)	Wo temp jis par air ko constant pressure par cool karne se vapour condense hona shuru hota hai (= p_v ke corresponding saturation temp). **Dew point depression** = DBT − DPT.
Absolute humidity	Unit volume moist air mein water vapour ka mass (kg/m^3). Kai books specific humidity ko bhi "absolute humidity" kehti hain.
Specific humidity (humidity ratio) ω	Per kg dry air water vapour ka mass: $\omega = 0.622\, p_v/(p - p_v)$ kg/kg d.a.
Relative humidity (φ)	Actual vapour mass / same DBT par saturated air ka vapour mass = $\varphi = p_v/p_s$ (%).
Degree of saturation (μ)	$\mu = \omega/\omega_s = \varphi\cdot(p - p_s)/(p - p_v)$. μ thoda sa φ se kam hota hai.

YAAD RAKHO Saturated air ke liye **DBT = WBT = DPT** aur φ = μ = 100%. Unsaturated air: DBT > WBT > DPT.

SOLVED EXAMPLE p = 1.01325 bar, DBT = 30 °C (p_s = 0.04246 bar), φ = 60%.
p_v = 0.6×0.04246 = 0.02548 bar; ω = 0.622×0.02548/(1.01325 − 0.02548) = **0.0160 kg/kg d.a.**
ω_s = 0.622×0.04246/(1.01325 − 0.04246) = 0.0272 ⇒ μ = 0.0160/0.0272 = **0.59**. DPT = saturation temp at 0.02548 bar ≈ **21.4 °C**.

UNIT IV Engineering Materials & Foundry

1. Classification of Engineering Materials

- **Metals:** (a) **Ferrous** – iron base: cast iron, wrought iron, steels; (b) **Non-ferrous** – Al, Cu, Zn, Pb, Sn and alloys (brass = Cu+Zn, bronze = Cu+Sn).
- **Non-metals:** polymers/plastics, ceramics (glass, refractory), rubber, wood; **Composites** (RCC, FRP, carbide tools).

Mechanical Properties

Strength: load sehne ki capacity bina failure (tensile, compressive, shear).

Elasticity: load hatane par original shape wapas.

Plasticity: permanent deformation bina tute (forging).

Ductility: tension mein wire mein kheencha ja sake (Cu, Al, mild steel).

Malleability: compression/hammering se sheet ban sake (gold, Al).

Toughness: fracture tak energy absorb karna (impact).

Hardness: scratch, indentation, wear resistance (Brinell, Rockwell).

Brittleness: bina deformation achanak tootna (cast iron, glass).

Stiffness: deformation resist karna (E high).

Resilience: elastic limit tak energy store (springs).

Fatigue: repeated/cyclic load mein yield se kam stress par failure.

Creep: high temp par constant load mein dheere-dheere permanent deformation.

2. Cast Iron

Cast iron = iron + **2–4.5% carbon** (theoretically 2.1–6.67%) + Si 1–3%, Mn 0.5–1%, S, P (impurities). Pig iron ko **cupola** mein remelt karke banta hai. Compressive strength high, tensile low, brittle, castability excellent, cheap.

Type	Structure / Making	Properties	Applications
Grey CI	Carbon **graphite flakes** ke roop mein (high Si, slow cooling); fracture grey	Good machinability, vibration damping, high compressive strength, brittle	Machine beds, engine blocks, pipes, manhole covers
White CI	Carbon **cementite (Fe_3C)** ke roop mein (low Si, fast cooling); fracture white	Very hard, wear resistant, brittle, not machinable	Grinding balls, crusher jaws; malleable CI ka raw material

Type	Structure / Making	Properties	Applications
Malleable CI	White CI ko long annealing (~900 °C) → temper carbon nodules	Ductile, tough, machinable	Pipe fittings, automobile parts, brackets
Nodular / SG / Ductile CI	Mg ya Ce add → **spheroidal graphite**	High strength, ductility (steel jaisi)	Crankshafts, gears, valves, pipes
Alloy CI	Ni, Cr, Mo, Cu add	Heat, corrosion, wear resistance	Cylinder liners, pump parts

3. Steels

Steel = iron + carbon **up to ~2%** (practically ≤1.5%) + Mn, Si etc. Carbon badhne par strength & hardness badhti hai, ductility & weldability ghat-ti hai.

Type	% Carbon / Composition	Properties	Applications
Low carbon / mild	up to 0.30% C (dead mild < 0.15)	Ductile, weldable, machinable	Wires, rivets, sheets, sections, bolts
Medium carbon	0.30–0.60% C	Stronger, harder, heat-treatable	Rails, axles, shafts, gears, crankshafts, connecting rods
High carbon	0.60–1.5% C	Very hard, wear resistant, less ductile	Cutting tools, springs, drills, files, chisels, razor blades
Stainless steel	Cr ≥ 10.5–12% (18/8 = 18 Cr, 8 Ni)	Corrosion resistant	Utensils, surgical tools, food & chemical plants
High speed steel (HSS)	18-4-1: 18% W, 4% Cr, 1% V	Red hardness (~600 °C tak hardness)	Drills, lathe tools, milling cutters
Other alloy steels	Ni, Cr-Ni, Hadfield Mn (12–14% Mn)	Toughness, wear resistance	Gears, rock crushers, rail crossings

4. Foundry (Casting)

> **DEFINITION** **Casting** = molten metal ko mould cavity mein daalkar solidify karwana jisse required shape mile. Jahan casting hoti hai use **foundry** kehte hain.

Basic steps: (1) Pattern making → (2) Mould & core making → (3) Melting (cupola for CI, crucible/induction furnace) → (4) Pouring (ladle se) → (5) Solidification & cooling → (6) Shake-out (mould todna) → (7) Fettling/cleaning (gates, risers, fins hatana) → (8) Inspection.

Equipment & tools: moulding boxes/flasks (**cope** – upper, **drag** – lower, **cheek** – middle), moulding board, shovel, riddle (sieve), rammer, strike-off bar, sprue pin, riser pin, vent wire, trowel, slick, lifter, draw spike, gate cutter, swab, bellows, ladle, cupola/crucible furnace, moulding machines (jolt, squeeze).

Patterns

Pattern = casting ka replica (thoda bada, allowances ke saath) jisse mould cavity banti hai.

Types:

- **Single piece (solid)** – simple shapes.
- **Split** – complex shapes; cope & drag mein half-half.
- **Loose piece** – projections alag nikaalne layak.
- **Match plate** – plate ke dono taraf; mass production.
- **Gated** – gates/runners attached; multiple small castings.
- **Sweep** – symmetric (bell, wheel) shapes, board ghuma kar.
- **Skeleton** – large castings, frame type.

Materials:

- **Wood** (teak, pine, mahogany) – cheap, easy, light; but moisture absorb, short life.
- **Metal** (Al, CI, brass) – durable, accurate; mass production; costly.
- **Plastic / epoxy** – light, smooth, moisture-proof.
- **Plaster of Paris** – intricate shapes.
- **Wax** – investment casting (pattern melt out).

Other types: cope & drag, follow board, segmental.

Allowance	Reason	Typical value
Shrinkage (contraction)	Solidification aur cooling par metal sikudta hai, isliye pattern bada	CI ~10 mm/m, steel ~20 mm/m, Al ~13 mm/m, brass ~15 mm/m
Machining (finish)	Machining se hatne wala extra metal	1.5–6 mm (size & metal par)
Draft (taper)	Pattern ko mould se bina damage nikaalne ke liye vertical faces par taper	~1–3° (external), inner surfaces par zyada
Distortion (camber)	U/V shape castings cooling par mudti hain, isliye pattern ulta mod ke banate hain	Shape dependent
Shake (rapping)	Pattern hilane se cavity badhti hai, isliye pattern chhota – **negative allowance**	Small

Moulding Sand – Types & Composition

Composition (typical green sand): silica sand (SiO_2) ~70–85% (refractoriness), **clay/bentonite** binder ~10–15% (strength), **water** ~4–8% (clay ko activate karta hai), additives (sea coal, cereal, wood flour, molasses) for surface finish & collapsibility.

Green sand: moist sand (silica + clay + water); sabse common, cheap.

Dry sand: green sand mould ko oven mein baked; strong, large castings.

Loam sand: ~50% clay; bade castings, sweep moulding.

Facing sand: pattern ke direct contact mein, fine, high quality.

Backing (floor) sand: purani reused sand, facing ke peeche.

Parting sand: clay-free dry sand; cope-drag ko chipakne se rokti hai.

Core sand: silica + oil binder (linseed); cores ke liye.

System sand: mechanised foundry mein poore mould ke liye.

Properties of good sand: permeability (gases nikal sake), refractoriness, cohesiveness (strength), adhesiveness, plasticity/flowability, collapsibility, reusability.

Mould Making (Green sand, two-box)

1. Moulding board par drag box rakho, drag half pattern rakho, facing sand phir backing sand bhar ke **ram** karo, strike off, vent holes.
2. Drag ko ulta karo, cope box rakho, cope half pattern lagao, **parting sand** chhidko.
3. Sprue pin aur riser pin lagao, cope mein sand bhar ke ram karo, vent karo, pins nikalo, pouring basin banao.
4. Cope uthao, pattern ko rap karke draw spike se nikalo, **gate** cut karo, mould repair, core rakho (agar hole chahiye).
5. Cope wapas rakh ke clamp karo; ab mould pouring ke liye ready.

Casting Defects – Causes & Remedies

Defect	Description / Cause	Remedy
Blow holes / gas porosity	Round smooth cavities; low permeability, excess moisture, hard ramming	Proper venting, moisture control, moderate ramming
Shrinkage cavity	Solidification shrinkage ko feed na milna; improper riser	Proper riser size/location, chills, directional solidification
Misrun	Metal poori cavity nahi bharta; low pouring temp, thin sections, slow pouring	Pouring temp & speed badhao, gating improve
Cold shut	Do metal streams milte par fuse nahi hote; low fluidity/temp	Higher pouring temp, better gating
Hot tear	Cooling ke time cracks; hard ramming, poor collapsibility	Collapsible sand/cores, uniform sections
Scab / Swell	Sand surface ukhad kar metal par; uneven/soft ramming → mould wall bulge	Uniform, proper ramming; good sand bonding
Sand / slag inclusion	Loose sand ya slag casting mein phansna	Clean mould, skim slag, strainer core
Mould shift / fins (flash)	Cope-drag misalignment; gap mein metal ki patli fin	Proper dowel pins, clamping, weights

UNIT V Fabrication (Welding) & Machine Tools

1. Welding & Weldability

DEFINITION **Welding** = do metal pieces ko heat (aur/ya pressure) se, filler metal ke saath ya bina, **permanently join** karna. **Weldability** = kisi metal ki wo capacity jisse use bina cracks/defects ke satisfactory weld kiya ja sake (low carbon steel – excellent; CI, high carbon steel – poor, preheat chahiye).

Classification of Welding Processes

Fusion welding (only heat, metal melts)	Pressure / Solid-state welding (heat + pressure)
Gas welding (oxy-acetylene); Arc welding – SMAW, TIG, MIG, SAW, carbon arc; Thermit welding; Electron beam, Laser beam	Forge welding; Resistance welding – spot, seam, projection, butt, flash; Friction welding; Ultrasonic, Explosive welding

Edge Preparation & Types of Joints



Edge preparation for butt joints (included angle of V ≈ 60–70°, root gap & root face)

Edge preparation zaruri hai taaki full penetration ho. Square (thin), single-V, double-V, single/double-U, J, bevel. **Types of joints: Butt** (same plane, end to end), **Lap** (overlap), **Tee** (90° T-shape), **Corner** (L-shape, box corners), **Edge** (parallel sheets ke edges). Welds: fillet weld (lap, tee, corner), groove weld (butt).

2. Gas Welding (Oxy-Acetylene)

Oxygen + acetylene (C_2H_2) jalane se ~**3200 °C** flame; filler rod aur flux se joint. **Equipment:** O_2 cylinder (black), C_2H_2 cylinder (maroon, acetone mein dissolved), pressure **regulators** with gauges, hoses (O_2 – black/green, C_2H_2 – red), **welding torch (blow pipe)** with nozzles, filler rods, flux, spark lighter, goggles, gloves.

Flame	$O_2 : C_2H_2$	Feature	Use
Neutral	≈ 1 : 1	~3200 °C, sharp inner cone, no chemical effect	Most metals – MS, CI, Cu, Al, SS
Carburising (reducing)	Excess C_2H_2	Feather (acetylene) zone, adds carbon	High carbon steel, hard facing, Monel
Oxidising	Excess O_2	Hottest (~3500 °C), short hissing cone	Brass, bronze

Techniques: Leftward (forehand) – thin plates (< 5 mm); Rightward (backhand) – thick plates. Gas cutting = oxy-acetylene se preheat + O_2 jet se iron oxidise.

3. Arc Welding

Electrode aur workpiece ke beech **electric arc** (~4000–6000 °C) metal pighlata hai. **Equipment:** power source – AC welding **transformer** ya DC **generator/rectifier** (OCV ~50–90 V, arc voltage ~20–40 V, current 50–500 A), electrode holder, welding cables, earth (ground) clamp, helmet/hand shield with dark glass, chipping hammer, wire brush, gloves, apron.

Electrodes

- **Consumable:** melt hokar filler banta hai – bare, lightly coated, **heavily coated (shielded)**. Mild steel core wire + flux coating.
- **Non-consumable:** carbon, graphite, **tungsten** (TIG) – pighalte nahi.
- **Flux coating ke kaam:** arc stabilise, shielding gas paida karna (oxidation se bachaav), **slag** banana (slow cooling, protection), alloying elements add karna, impurities hatana.

Polarity (DC welding)

Polarity	Connection	Heat distribution	Use
DCSP / DCEN (straight)	Electrode −ve, work +ve	~2/3 heat workpiece par → deep penetration	Thick plates, steel
DCRP / DCEP (reverse)	Electrode +ve, work −ve	~2/3 heat electrode par → faster electrode melting, shallow penetration	Thin sheets, non-ferrous, coated electrodes
AC	Har half-cycle badalti	Equal heat	General MS welding

Arc Welding Processes

SMAW (MMAW, stick): flux-coated consumable electrode, manual; general fabrication.

TIG (GTAW): non-consumable tungsten electrode + argon/helium shield, alag filler; Al, SS, thin sheets, high quality.

MIG (GMAW): continuously fed consumable wire + inert gas (Ar/CO_2); fast, semi-automatic.

SAW (Submerged arc): arc granular flux ke neeche chhupa; thick plates, long straight seams, automatic.

Carbon arc: carbon electrode, filler alag. **Resistance spot:** I^2Rt heat + pressure; sheet metal, car bodies.

Gas vs Arc: gas – flame ~3200 °C, wide HAZ, more distortion, thin sheets, slow, portable (no electricity); arc – ~4000–6000 °C, narrow HAZ, thick plates, faster & stronger joint, needs power supply.

4. Soldering & Brazing

Point	Soldering	Brazing	Welding
Filler melting temp	< 450 °C; base metal not melted	> 450 °C but < base metal; base not melted	≈ base metal m.p.; base melts
Filler	Soft solder Sn-Pb (60/40), lead-free Sn-Ag-Cu	Brass (Cu-Zn, spelter), silver alloy	Same as base metal
Flux	Rosin, $ZnCl_2$, NH_4Cl	Borax	Electrode coating / flux
Strength / use	Weakest; electronics, PCB, wires	Medium; carbide tool tips, pipes, radiators, cycle frames	Strongest; structures

Joining capillary action se hota hai (soldering/brazing). **Braze welding** mein filler capillary se nahi, gas welding jaise V-groove mein bhara jata hai (CI repair).

5. Welding Defects – Causes & Remedies

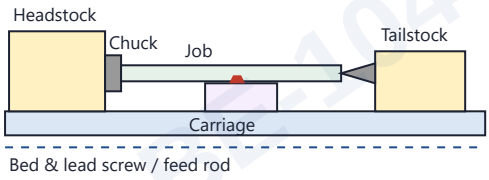
Defect	Cause	Remedy
Porosity	Moist/dirty electrode or surface, gas entrapment, long arc	Dry electrodes, clean surface, correct arc length

Defect	Cause	Remedy
Slag inclusion	Slag not removed between passes, low current	Chip & brush each pass, proper current/angle
Incomplete fusion / penetration	Low current, high speed, poor edge prep, small root gap	Higher current, slower speed, proper V & gap
Undercut	Excess current, long arc, wrong electrode angle	Reduce current, correct angle & arc length
Cracks	High carbon/hardenable steel, rapid cooling, restraint	Preheat, post-heat, low-hydrogen electrode
Spatter / Overlap	Too high current, long arc / slow travel, low current	Correct current, arc length & travel speed
Distortion	Non-uniform heating & cooling	Clamping, tack welds, balanced sequence

6. Machine Tools

Machine tool = power driven machine jo cutting tool se workpiece ka extra metal (chips) hata kar required shape, size, finish deta hai. Basic need: work aur tool ko hold karna, relative motion (cutting + feed) dena. **Cutting speed V = πDN/1000 m/min** (D in mm, N in rpm). Example: D = 50 mm, N = 400 rpm ⇒ V = **62.8 m/min**. Lathe machining time T = L/(f·N) min.

(a) Lathe – "Mother of all machine tools"



Centre lathe – main parts

Parts: Bed (guideways), **Headstock** (spindle, gear box, chuck), **Tailstock** (dead centre, drill holding), **Carriage** = saddle + cross slide + **compound rest** + tool post + apron, **feed rod**, **lead screw** (threading), legs.

Specification: height of centres, **swing over bed** (max dia), **distance between centres** (max job length), bed length, spindle bore, no. of speeds, motor power. e.g. "Lathe 1500 × 300 mm".

Tools: single-point cutting tools (HSS, carbide-tipped) – turning, facing, parting, threading, boring tools, knurling tool. **Operations/Applications:** facing, plain & step turning, **taper turning** (compound rest, tailstock set-over, taper attachment, form tool), thread cutting, drilling, boring, reaming, knurling, chamfering, grooving, parting-off. Shafts, bolts, pulleys, cylindrical jobs.

(b) Drilling Machine

Types: portable, **sensitive (bench)** – small holes up to ~15 mm, hand feed; **upright (pillar/column)** – medium jobs; **radial** – heavy jobs, arm column ke around ghoomta hai; gang, multi-spindle. **Parts:** base, column/pillar, table, radial arm (radial), drill head, spindle, chuck/sleeve, feed handle, motor. **Specification:** max drill dia, spindle travel, table size, distance spindle to column/table, (radial: arm length, column dia), speeds, power. **Tools:** twist drill (point angle **118°**, HSS), reamer, counterbore, countersink, tap. **Operations:** drilling, reaming, boring, counterboring, countersinking, spot facing, tapping, trepanning.

(c) Shaper

Single-point tool **ram** par reciprocate karta hai; work table par fixed, har return stroke ke baad feed. **Forward stroke = cutting**, return idle – **quick return mechanism** (crank & slotted lever) se return fast (cutting : return time ≈ 3 : 2). **Parts:** base, column/body, ram, tool head (clapper box, tool post, down-feed screw), table, cross rail, saddle, drive mechanism. **Specification: max stroke length** (e.g. 600 mm), table size, max travel, power, ratio of cutting to return time. **Applications:** flat horizontal, vertical, inclined surfaces, keyways, slots, grooves, splines.

(d) Milling Machine

Multi-point rotating cutter, work table par feed hota hai; high metal removal, accurate. **Types:** column & knee – **horizontal**, **vertical**, **universal** (table swivel, helical milling); planer type, bed type. **Parts:** base, column, **knee** (vertical move), saddle (cross), **table** (longitudinal), spindle, arbor, overarm, dividing (indexing) head. **Specification:** table working surface, longitudinal/cross/vertical travel, spindle speeds & feeds, power, spindle nose taper. **Cutters:** plain (slab), side & face, end mill, face mill, slot/slitting saw, angle, form, gear cutter. **Up milling** (conventional – cutter rotation feed ke opposite) vs **down milling** (climb – same direction, better finish, needs backlash eliminator). **Operations:** plain/face/angular milling, slotting, keyway, gear cutting, straddle & gang milling.

(e) Grinding Machine

Rotating **abrasive wheel** se bahut kam metal hata kar **high accuracy & fine finish** (hard materials bhi). **Types: surface** (flat), **cylindrical** (external – between centres), **internal** (holes), **centreless** (no centres; regulating wheel – pins, rods mass production), tool & cutter grinder, bench/pedestal grinder. **Parts:** base, table, work head, tailstock, wheel head, wheel guard, coolant system. **Specification:** max dia & length of job, wheel size, table size, power. **Wheel:** abrasive – Al_2O_3 (steels), SiC (CI, non-ferrous, carbide), diamond, CBN; grain size, grade (A soft–Z hard), structure, bond (V vitrified, B resinoid, R rubber, E shellac, S silicate, M metallic). Marking e.g. **51 A 46 K 5 V 23**. **Dressing** = clogged wheel ko sharp karna; **truing** = wheel ko shape/round mein lana.

> **EXAM TIP** Har machine: principle → types → labelled sketch of parts → specification → operations. Neat diagram = extra marks.